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Research paper

Deep neuro-neutrosophic systems for detection of online sexism

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ABSTRACT

This paper proposes a deep neuro-neutrosophic system, a novel framework that combines DNN with the Neutrosophic Inference System (NIS) to improve prediction capacity. With the increasing usage of online social media platforms and anonymity, there is a multifold increase in sexism and hateful comments online. The proposed framework is utilized on the SemEval-2023 Task 10 dataset (with three subtasks) for the explainable detection of online sexism.

The proposed approach for subtask A (to detect if a particular comment) is the pre-trained RoBERTa (Robustly Optimized BERT Approach) base model combined with a NIS based on Single-Valued Neutrosophic Set (SVNS) to classify the comments considering ambiguity. The Bidirectional Long Short-Term Memory (BiLSTM) model and suitable NIS based on refined neutrosophic sets were combined for subtasks B and C (to classify into the type of sexism). The proposed system provided higher accuracy than the traditional classification method and other existing models in all subtasks; it obtained 93.1% accuracy in subtask A, 91.55% in subtask B, and 97.98% in subtask C, compared to deep neural network (DNN) models that obtained 91.6% accuracy in subtask A, 78.74% in subtask B, and 78.77% in subtask C.

The proposed framework can be used in various real-world scenarios that involve content moderation, including detecting online sexism spread through social media platforms. This research concludes that deep neuro-neutrosophic systems improve on existing deep learning models and are good at detecting online sexism and producing accurate results.

1. Introduction

Online sexism negatively impacts female users by creating malicious online experiences and perpetuating inequalities forever. Many automated tools for detecting sexist comments have limited insight, categorizing them into sexist or non-sexist. Sexism and misogyny are prevalent mindsets on Online Social Networks (OSN), strengthening ancient patriarchal attitudes of name-calling, objectifying appearances, and stereotyping gender roles (Gasparini et al., 2022). Increasing internet access has also facilitated the spread of sexism online in the form of harassment and discrimination based on gender. Natural Language Processing (NLP) models can efficiently analyze large amounts of text, making it suitable for monitoring potentially sexist content.

Datasets given in sEXism Identification in Social neTworks (EXIST 2023) (Gasparini et al., 2022) and SemEval 2023 Task 10: Explainable Detection of Online Sexism (Kirk et al., 2023) were curated by researchers for online sexism detection in recent years. SemEval 2023 Task 10 on online sexism (Kirk et al., 2023; Ansari et al., 2023) contains three subtasks, as summarized in Fig. 1.

It provides benchmarks for detecting sexist content in textual data, evaluated with a macro-average F1 score to account for class imbalance. SemEval-2023 Task 10 dataset (Kirk et al., 2023) focuses on identifying and categorizing online sexism across three subtasks, providing crucial benchmarks for model evaluation, for binary classification (subtask A), which distinguishes sexist from non-sexist text, standard metrics like accuracy and F1-score serve as benchmarks, with models like logistic regression providing initial baselines (Kirk et al., 2023). However, the uneven nature of sexist content makes this task difficult. Thus, transformer-based models such as Bidirectional Encoder Representations from Transformers (BERT) and Robustly Optimized BERT Pre-training Approach (RoBERTa) can capture complex language patterns substantially better.

In subtask B, multi-class classification is employed on the resulting dataset, which identifies four classes of sexism: threats/incitement, derogation, anger, and prejudiced conversations. The performance metrics are macro and weighted F1 measures, but each of these classes poses a set of different challenges. Subtask C extends fine-grained

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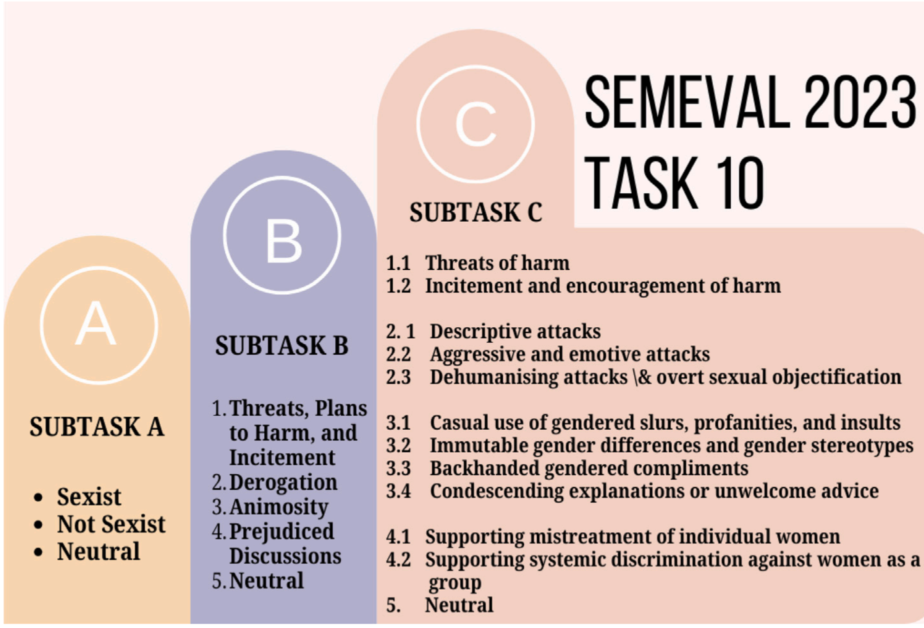

Fig. 1. Categorization of the comments across subtasks A, B, and C of the SemEval 2023 Task 10 on online sexism detection.

classification into 11 detailed vectors of sexism. Benchmarks for these tasks are based on per-class precision, recall, and F1 scores, with subcategories such as backhanded compliments or even condescending counsel expected to test even sophisticated models. These classification tasks require a granular representation to enhance the performance of the model.

Deep learning models frequently struggle with the ambiguities and uncertainties that are natural in human language. While they have been proven to be powerful in capturing patterns, they often struggle with human language's ambiguity. Deep learning models benefit from incorporating neutrosophic principles to handle these difficulties better. To enable models to provide more complex predictions, neutrosophic frameworks can offer extra levels of representation that capture the varying degrees of truth, falsity, and uncertainty in textual input.

Neutrosophy is a field that examines neutralities and their interactions with different perspectives, introduced by Smarandache (1999). It extends fuzzy logic to include a neutral or indeterminate state along with the truth and falsehood. Neutrosophic logic allows for a more nuanced understanding of uncertain, vague, or contradictory phenomena across various domains. By incorporating neutrality, neutrosophy provides a comprehensive problem-solving approach and has been applied in solving real-world problems as reviewed in Peng and Dai (2020).

To implement neutrosophic philosophy to real-world and engineering problems, Wang et al. (2010) proposed Single Valued Neutrosophic Sets (SVNSs). It was further developed by Smarandache (2013) to refined neutrosophic set, with more membership functions for better granular representation of data; refined neutrosophic sets provide a more sophisticated approach to understanding complex information by building upon the fundamental concepts of neutrosophy (Deli et al., 2015; Broumi and Deli, 2016). Specifically, refined neutrosophic sets establish a mathematical framework for nuanced quantification of indeterminate, incomplete, or inconsistent data. These sets incorporate various degrees of truth, falsehood, and indeterminacy, which facilitate a granular representation of uncertain information (Deli, 2016).

Neutrosophic logic, with its ability to model indeterminacy, can address this challenge. However, its application in NLP, especially in social problems, has not yet been explored. This paper is motivated by the need to bridge that gap, using neutrosophic principles alongside deep learning to improve online detection and classification of subtle and complex sexist content.

In NLP and addressing online sexism, neutrosophy can help to improve system accuracy by acknowledging ambiguities and modeling indeterminate language states to handle subtle sexist language better. Within the domain of online content moderation, neutrosophic sets can improve NLP models' ability to comprehend and categorize subtle forms of language expression, such as sarcasm and irony (Sharma et al., 2021a,b). This approach could yield improvements in detecting and classifying sexist content, adding to the effectiveness of automated moderation systems. Despite the chance that this technique can contribute to more precise analysis and informed decision-making across various fields by facilitating a richer assessment of available information, it has not been researched or implemented till now.

This paper proposes deep neuro-neutrosophic systems as a generalization of deep neuro-fuzzy systems. The proposed framework facilitates integration that utilizes both DNN and neutrosophy. Many researchers have previously combined deep learning models with neutrosophy and obtained interesting results (Sharma et al., 2021b; Shitaya et al., 2025; Mostafa et al., 2021).

Contributions of the paper:

- Introduced refined neutrosophic sets with variable representation for subtask A (with SVNS (3 memberships)), subtask B (refined neutrosophic sets with five memberships), and subtask C (refined neutrosophic set with 12 memberships).
- This paper proposes deep neuro-neutrosophic systems that combine DNNs and NIS that can be used for various classification tasks with little fine-tuning.
- For each subtask, a different system has been proposed, implemented, and compared against existing models.
- The proposed systems improve the detection accuracy in all three subtasks. It achieved an accuracy of 93.1% in subtask A, 91.55% in subtask B, and 97.98% in subtask C.

This deep neuro-neutrosophic systems method exploits neutrosophic sets' ability to handle bias and uncertainty and deep learning's pattern recognition skills to produce more accurate and dependable sexist language detection. The NIS model incorporates degrees of sexism and various categories of sexism to capture overt and covert types of sexism. Refined neutrosophic sets can handle this representation of these fine-grained classifications.

The rest of the paper is organized as follows: Section two provides a comprehensive survey of the previous research. Section three introduces deep neuro-neutrosophic systems, their related concepts, and models. The methodology, including the dataset, preprocessing steps, feature extraction techniques, and the architecture of the proposed deep neuro-neutrosophic systems, are given in section four. It also explains the integration of NIS using SVNS and refined neutrosophic sets in the model to handle biases and uncertainties. Section five deals with experimental results and discussions related to the tasks and result analysis. It also highlights the strengths and limitations of the proposed framework and the study's theoretical and practical implications. The last section concludes the study and discusses future research possibilities.

2. Literature survey

NLP models can offer a powerful solution for detecting online sexism in comments, posts, or tweets. The review paper (Khurana et al., 2022) traverses through all the components of NLP, explaining each classification. It also discusses the applications, including text classification and information extraction. It explains the state-of-the-art (SOTA) models and the available datasets in NLP. Several frameworks demonstrate the capability of NLP in real-world applications. For instance, the Datalyzer system (Alvarado et al., 2024) showcases how NLP can bridge the gap between human language and structured data systems. It enables non-technical users to formulate natural language queries converted into structured queries for data retrieval, visualization, and real-time predictions. This illustrates NLP's potential for making complex data systems more accessible.

CivilityCheck (Bonthu et al., 2024) further demonstrates this by utilizing logistic regression techniques to process and model text data that is obtained from Twitter. It processes the data using standard NLP techniques like cleaning, tokenization, and stemming to classify tweets into hate speech, offensive language, and neutral categories. The framework achieves high accuracy and scalability, underlining the importance of thorough text preprocessing in effective classification.

A review of the development and application of Neural Language Models (NLM) used to improve existing NLP systems is given in Chen (2023). These NLP systems utilize deep learning models, which are neural networks, to solve language-related problems like sequence prediction and text classification effectively. The models look at shortcomings in traditional statistical techniques, such as n-grams, which consider only word similarities and fail to capture long-term dependencies in language.

The increased usage of hate speech in online platforms has underlined the need to design effective detection methodologies primarily based on NLP techniques. Recent works have demonstrated how SOTA models, like GPT-2, can effectively detect hate speech by using deep learning and NLP to classify subtler language cues unaccounted for by traditional algorithms. Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) models have been particularly noted for their ability to read sequential data and capture the meaning through temporal context that is both current and historical in users' tweets. While RNN networks can provide a rudimentary framework for understanding the temporal and widespread relationships of user expressions, they struggle with long-term dependencies due to gradient problems in learning. Alternatively, the LSTM networks were designed to store information in long sequences, providing even more flexibility when interpreting potentially nuanced emotional expressions in tweets.

The work by Mohammadi et al. (2024), presented for the EXIST 2023 tasks, uses a multi-model NLP-based approach for hate speech and abusive language detection. It deals with DistilBERT, BERT, and XLM-RoBERTa models on the EXIST 2023 dataset. The method used includes data preprocessing, additional datasets, and model building. The authors also highlight future research directions, including refining annotator reliability and improving methods for handling imbalanced data. In Maqbool (2024), machine learning models are combined with deep learning models such as BERT for the shared task of sexism detection in social media posts as part of EXIST 2023. The probability-based techniques applied in the random forest had remarkable outcomes where they outperformed the detection of sexist content in comparison with deep learning models, especially with the multilingual BERT. However, deep learning models generally performed better than the random forest in evaluation metrics. They thus represented stronger performance and better multilingual data handling, especially in both English and Spanish.

An increase in sexist content on OSN platforms has driven the need for automatic detection of tweets or posts and classification as suggested in Abburi et al. (2024). It explores the detection of sexist content in a Twitter or Gab post. The sexist comments, once detected, are then divided further into classifications and types of sexism they come under. The proposed neural model uses RoBERTa as well as linguistic features like Hurtlex, Empath, and Perspective API. The framework includes a knowledge module that uses emoticons and hashtags.

Similarly, the work by Hamdy et al. (2021) focuses on the role of neural models in detecting offensive language in OSN. It reviews offensive language classification and compares various machine learning and deep learning methods that would apply. It also talks about context because, in many cases, offensive words may rely on subtle linguistic cues. The study also identifies key datasets and studies the performance of neural networks in dealing with complex textual data, particularly with transformer-based architectures such as BERT.

Techniques to detect offensive language in multilingual social media posts were dealt with in Singh and Mishra (2022). Specifically, this research applies the methods of Multilingual Offensive Language Detection, where deep learning and NLP are incorporated features for selection and classification. One of the main features of the study is the fuzzy-based Convolutional Neural Network (CNN) method for feature selection, and the use of an ensemble architecture of a BiLSTM model hybridized with Naïve Bayes and SVM for classification. The methodology is equally efficient for offensive content in multiple languages.

Deep learning techniques are integrated with neutrosophy (Sharma et al., 2021b,a) to improve sentiment analysis and especially predict neutralities in offensive language identification datasets. The authors have identified an important gap in current research, which leaves out neutralities in favor of a simple classification between offensive and non-offensive content. They use the forefront models of the field, namely BiLSTM, BERT, ALBERT, RoBERTa, and MPNet, which boast feature extraction from intermediate layers, and they apply clustering techniques to classify the sentiments into SVNS. This innovative approach not only quantifies sentiments more effectively but also aids in the extraction of neutral tweets from datasets majorly labeled as offensive or non-offensive.

Neutrosophic sets, developed by Smarandache in the early 2000s, have received much attention for resolving uncertainty and indeterminacy in various domains. Unlike fuzzy sets, which only capture membership degrees, neutrosophic sets include truth (T), indeterminacy (I), and falsity (F) values, and refined neutrosophic sets also provide a more nuanced framework for dealing with imprecise data. Wang et al. (2010) expanded the application of neutrosophic sets in decision-making difficulties, diagnosed medical conditions, and analyzed many criteria. Their utility in handling real-world uncertainty has been demonstrated in AI, computer science, and network optimization.

Various refined neutrosophic sets like double valued neutrosophic sets (Kandasamy, 2018), bipolar neutrosophic sets, Triple Refined Indeterminate Neutrosophic Sets (TRINSs) (Kandasamy and Smarandache, 2016) have been defined, and applied in solving various real-world problems like sentiment analysis (Mishra et al., 2020), psychology (WB et al., 2020), medicinal field (Hashim et al., 2020; Dutta and Borah, 2023) and social problems (Zhao, 2025; Luo and Wang, 2025).

A thorough analysis of neutrosophic logic and the extensions of neutrosophic logic is carried out in Pramanik (2022), which includes

SVNS and multi-valued neutrosophic sets, which are extensions of fuzzy and intuitionistic fuzzy sets. Neutrosophic and fuzzy logic are compared and contrasted, and the authors asserted that the rationale for neutrosophic logic extends beyond that proposed for fuzzy and intuitionistic fuzzy logic, particularly in voting scenarios that require decision-making contexts where fuzzy logic does not adequately capture all the required dimensions or uncertainty. Similarly, SVNS uses trapezoidal and triangular membership functions to help account for the stochastic nature of systems. At the same time, the authors suggested SVNS will provide flexibility and accuracy amid the same limits established with traditional models.

A novel approach with an amalgamation of sentiment analysis, neutrosophic set, and multi-attribute decision-making meant to rank alternative products based on online consumer reviews was proposed (Awajan et al., 2021). The method, termed Neutro-VADER, expands the basic VADER sentiment tool. It accommodates uncertainty by reflecting the positive, neutral, and negative scores as degrees of neutrosophic numbers, which indicate truth, indeterminacy, and falsity developing on the work by Kandasamy et al. (2020), which analyzed #Metoo tweets using SVNS. This method ranks alternatives and exhibits better handling of neutral data compared to traditional approaches for sentiment analysis techniques.

A new opinion mining model for social media is presented in Essameldin et al. (2022), specifically for handling perspectivism and ambiguity in sentiment classification. The study combines social network analysis and Artificial Neural Networks (ANN) to rank users and incorporate their influence into the sentiment analysis, using TextBlob for polarity scoring. It compares three uncertainty classifiers: Type-1 fuzzy logic, Type-2 fuzzy logic, and neutrosophic logic. Results indicate that neutrosophic logic yields the highest accuracy of handling ambiguity in data, mirroring the actual opinion dynamics. Voskoglou (2023) deals with the limitations of fuzzy sets by introducing neutrosophic triplets, which include truth, indeterminacy, and falsity as decision criteria. This gives way to more flexible decision-making in situations wherein the data given may be conflicting or incomplete. Voskoglou's work offers a holistic framework for decision-making whenever there is a failure of traditional binary logic. Mallik et al. (2023) explores how recommender systems have been increasingly used for customizing applications so that a user gets a more personalized experience. The paper mainly discusses introducing neutrosophy into recommendation systems to improve them. It was found that the neutrosophic method had a comparatively higher accuracy than traditional deep learning methods.

SVNS is used in Yuan and Zhang (2022) for optimal modification of the similarity measures based on the weight of true, uncertain, and non-true value membership degrees. The investigation reveals that increasing the weight of the true value membership degree results in a 2%–3% increase in recommendation accuracy. This strategy is relevant to our research on detecting online sexism using NLP, as SVNS can handle uncertainty and subjectivity in sexist language. In targeting the true value membership degrees, such as modifications of the other types of membership values, we can elevate our model's capacity to sense and, thus, spot unsaid sexist content effortlessly. This move could raise the model's performance and awareness by grasping subtler forms of online sexism that usually go undetected by binary approaches.

Despite the significant theoretical advancements in neutrosophic logic and its application in various domains, several key research gaps remain, particularly in the practical application of neutrosophic logic in NLP and other AI-driven fields. A combination of neutrosophic logic and deep learning has been carried out extensively in image processing by several researchers (Wajid et al., 2024; Tolba and Talal, 2024; Abdullah, 2024; Mostafa et al., 2021; El-Shahat and Tolba, 2024). However, there is only limited work that combines neutrosophy with NLP.

The NLMs' ability to estimate the probability distribution of the next word based on the sequence of words and the surrounding context makes them especially applicable to the proposed research on detecting online sexism. Here, NLM can be utilized to handle messages, find the patterns of sexism, and categorize input as sexism or nonsexism. In the same vein, one of the most recent techniques in detecting linguistic nuances is the application of NLMs together with SVNS (Sharma et al., 2023, 2021b), which consequently enhances the processing of truth, indeterminacies, and falsities that are part of online discourses.

Research Gap: NLP models such as BERT, RoBERTa, and DeBERTa have successfully detected online sexism and offensive language, but the use of SVNS or neutrosophic sets in these domains has received less attention. SVNS allows for greater flexibility in dealing with uncertainty and vagueness, which are required when classifying ambiguous or context-dependent statements (Kamath et al., 2022). Existing research, however, takes SVNS and neutrosophic logic for granted to represent inexactness inside deterministic or probabilistic frameworks without thoroughly examining its potential to deal with subjective input. Because neutrosophic sets can handle ambiguous linguistic expressions between explicitly non-sexist and sexist content, we can incorporate their use into state-of-the-art pre-trained models (e.g., RoBERTa or BiLSTM) to construct richer models for offensive content classification. Models that use neutrosophic sets to manage uncertainty can often be more reliable.

Neutrosophic sets can be applied broadly to better handle ambiguity and uncertainty in observational descriptions; their combination with existing NLP models could lead to improved categorization performance in online content moderation, patents, and legal documents. Also, refined neutrosophic sets can aid in improving the representation of ambiguity and uncertainty present in natural languages. The integration of fuzzy logic and neural systems has been generally grouped as neuro-fuzzy systems; in the same vein, in this paper, we propose the integration of neutrosophic logic and neural systems as neuro-neutrosophic systems.

Integrating DNNs with neutrosophic sets, especially in applications such as online sexism detection, can create a more accurate classification system. While advanced NLP models such as DeBERTa, BERT, and RoBERTa have improved text analysis, they lack integrated strategies for handling ambiguity, particularly in sensitive domains. The proposed system addresses this gap.

Despite the progress of neuro-fuzzy systems in addressing real-world problems, the intersection of neutrosophy and deep learning remains largely unexplored. This paper proposes deep neuro-neutrosophic systems as a comprehensive framework for merging neural networks and neutrosophy.

3. Deep neuro-neutrosophic systems

3.1. Neutrosophic sets

Neutrosophy is a contemporary extension of philosophy that examines the range of neutralities and their interactions. The formal conceptualization of neutrosophy is defined as:

Definition 1. Given $\langle B\rangle$ to be an idea, or theory, or proposition, or event, or entity, by $\langle not\ B\rangle$ we consider what is not $\langle B\rangle$, and by $\langle Anti\ B\rangle$ the opposite of $\langle B\rangle$. Also, $\langle Neut\ B\rangle$ means what is neither $\langle B\rangle$ nor $\langle Anti\ B\rangle$, i.e., neutrality in between the two extremes. Fundamentally, any idea $\langle B\rangle$ has T as truth, I as indeterminate, and F as false, where $T, I, F \subset \|^{-}0, 1^{+}\|$

If given a sexist tweet B, using neutrosophic set in this form, it can be represented using $\langle T_B, I_B, F_B\rangle$ and given values like $T_B = 1.12$ or even greater. Given this nature of the neutrosophic set, it was not feasible to implement the actual definition of a neutrosophic set in real-world problems; hence Wang et al. in 2010 (Wang et al., 2010) introduced the idea of SVNS. The original definition of SVNS bounds the membership values to [0,1], as given in Wang et al.

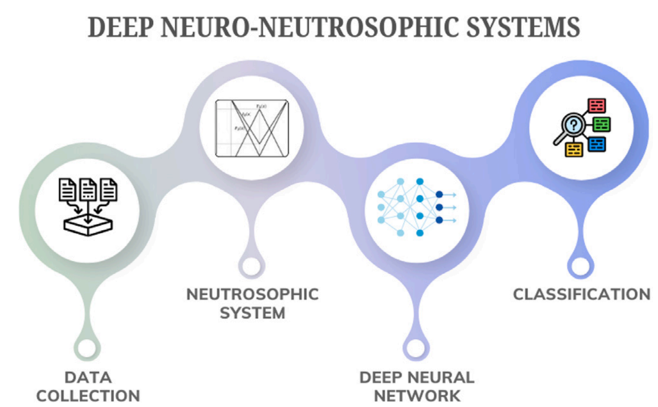

Fig. 2. A sequential deep neuro-neutrosophic systems that incorporate a neutrosophic system that handles the input and the resultant is then utilized by some DNN-based model.

Definition 2. Let X be a space of objects, with an element in X denoted by x. An SVNS A in X is characterized by three membership functions, namely, truth-T_A, indeterminacy I_A, and falsity F_A. For each point x in X, $T_A(x), I_A(x), F_A(x) \in [0, 1]$.

This definition of SVNS cannot be directly used to map a tweet or a comment as sexist or not. Hence, the definition of SVNS, altered to suit the purpose of classifying sexist tweets, will be as follows:

Definition 3. An SVNS of a comment C, in the context of online sexism detection, will be defined as a C ∈ U (universal set) is $C = \{(t, S(t), A(t), NS(t)) : t \in U, S(t), A(t), NS(t) \in [0, 1], 0 \le S(t) + A(t) + NS(t) \le 3\}$, where $S(t), A(t), NS(t)$ are the degrees of sexist (or membership), indeterminacy (or neutrality or ambiguous) and not-sexist (or non-membership) of $t \in A$ respectively, called the neutrosophic components of x.

Sample tweets from the subtask A that are considered for illustrative purposes are given below:

"You have to realize that in today's universities, 80% of these ẅomenäre fat, sjws/feminists, and in no way more than a 3–4. Of the other 20%, maybe 30% are decent looking, fuckable women who won't scream rape after you alpha widow them".

This tweet, which is taken from the dataset, is used to show how SVNS will handle the tweet and define memberships, to show the sexist, non-sexist, and neutral membership functions, according to subtask A. Consider this sample tweet as some j, to represent the results of the tweet in SVNS, three memberships will be used. $\langle S_j, N_j, NS_j\rangle = \langle 0.92, 0.14, 0.07\rangle$ is the value of the membership functions, which implies that the sexist membership value is 0.92, the non-sexist membership value is 0.07, and the neutral membership value is 0.14.

Let us use another sample tweet k:

Women always bitching about being a mom yet they keep having them.

The SVNS representation of this tweet with three memberships will be $\langle S_k, N_k, NS_k\rangle = \langle 0.85, 0.4, 0.35\rangle$, which implies that the sexist membership value is 0.85, the non-sexist membership value is 0.35, and the neutral membership value is 0.4.

SVNS can be used only to represent positive, negative, and neutral membership functions, to overcome this limitation, neutrosophic sets were further developed into refined neutrosophic sets of several types. To handle the memberships related to Ekman emotion, refined neutrosophic sets were defined in Sharma et al. (2023). Similarly, refined neutrosophic sets are defined here to cater to the classification that needs to be performed for subtask B.

While considering subtask B, a text can be classified into

1. Threats, Plans to Harm, and Incitement
2. Derogation
3. Animosity
4. Prejudiced Discussions
5. Neutral

making it have five possible classes. To closely represent the classification probabilities, we define RNSs with five memberships, similar to TRINSs from Kandasamy and Smarandache (2016).

Definition 4. Consider every text $t \in T$, T a space of all text vectors. A TRINS $R \in T$ is characterized by Threats $T_R(t)$, derogation $D_R(t)$, Animosity $A_R(t)$, Prejudice $P_R(t)$, and Neural $N_R(t)$, membership functions $\forall t \in T$, where $E_R(t) \in [0, 1]$; (E denotes any of the sexist categories) and $0 \le \sum E_R(t) \le 5$. Therefore, the TRINS of a text can be represented as five-valued refined neutrosophic sets given in the equation.

$$S = \{\langle j, T_R(j), D_R(j), A_R(j), P_R(j), N_R(j)\rangle j \in J\} \tag{1}$$

For the previously provided tweet example, the following shows how the TRINS will be represented,

$$\begin{aligned} TRINS_j &= \{\langle T_R(j), D_R(j), A_R(j), P_R(j), N_R(j)\rangle\} \\ &= \langle 0.38, 0.72, 0.63, 0.49, 0.15\rangle \end{aligned}$$

While handling subtask C, the following classification needs to be done: under each category of subtask B, new subcategories are defined for subtask C, this classification is listed as follows:

1. **Threats, Plans to Harm, and Incitement:**
   - 1.1 Threats of harm:
   - 1.2 Incitement and encouragement of harm:
2. **Derogation**
   - 2.1 Descriptive attacks:
   - 2.2 Aggressive and emotive attacks:
   - 2.3 Dehumanizing attacks & overt sexual objectification
3. **Animosity**
   - 3.1 Casual use of gendered slurs, profanities, and insults:
   - 3.2 Immutable gender differences and gender stereotypes:
   - 3.3 Backhanded gendered compliments
   - 3.4 Condescending explanations or unwelcome advice
4. **Prejudiced Discussions:**
   - 4.1 Supporting mistreatment of individual women
   - 4.2 Supporting systemic discrimination against women as a group
5. Neutral

Similarly, Intricate Refined Neutrosophic Sets (IRNSs) are defined to aid in the classification of subtask C to accurately capture the granularity of the data and categorize the tweet accordingly.

Definition 5. Consider a space of text with every text $t \in T$. An IRNSs $R \in T$ is characterized by Threats $T_{1_R}(t)$, $T_{2_R}(t)$, derogation $D_{1_R}(t)$, $D_{2_R}(t)$, $D_{3_R}(t)$, Animosity $A_{1_R}(t)$, $A_{2_R}(t)$, $A_{3_R}(t)$, $A_{4_R}(t)$, Prejudice $P_{1_R}(t)$, $P_{2_R}(t)$ and Neural $N_R(t)$, membership functions $\forall t \in T$, where $E_R(t) \in [0, 1]$; (E denotes any of the sexist categories) and $0 \le \sum E_R(t) \le 12$. Therefore, the RNS of a text can be represented as twelve-valued refined neutrosophic sets given in the equation.

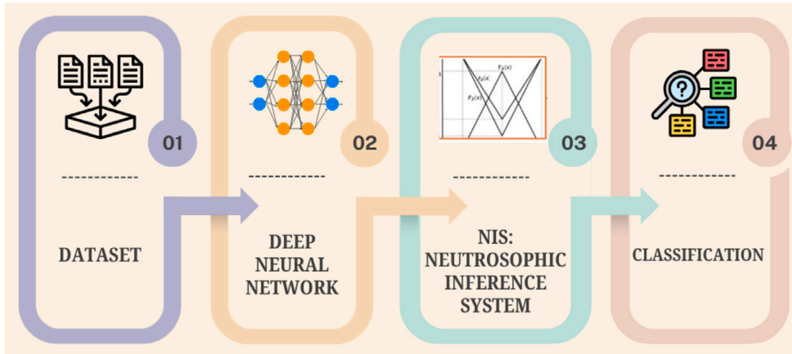

Fig. 3. The generic architecture of sequential deep neuro-neutrosophic systems, which incorporates a DNN first, and the features extracted are utilized for the working of the NIS that follows and performs the classification.

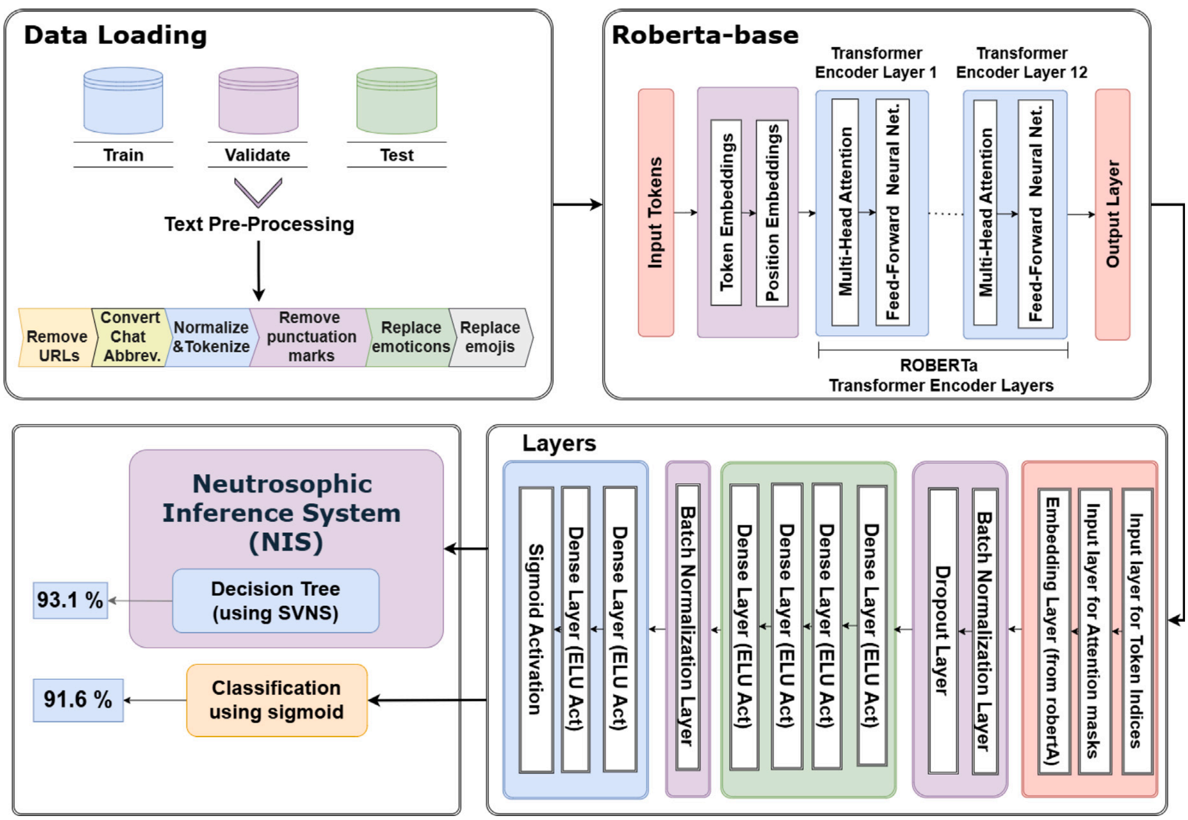

Fig. 4. The system architecture of the deep neuro-neutrosophic system that is based on RoBERTa and then integrated with the NIS-based decision tree for the classification task of subtask A.

Considering the same example tweet while classifying for subtask C. The membership values with description is as given T_{1_A} (threats of harm) = 0.47; T_{2_A} (incitement and encouragement of harm) = 0.35; D_{1_A} (descriptive attacks) = 0.77; D_{2_A} (aggressive and emotive attacks) = 0.72; D_{3_A} (dehumanizing attacks & overt sexual objectification) = 0.77; A_{1_A} (casual use of gendered slurs, profanities, and insults) = 0.68; A_{2_A} (immutable gender differences and gender stereotypes) = 0.62; A_{3_A} (backhanded gendered compliments) = 0.41; A_{4_A} (condescending explanations or unwelcome advice) = 0.49; P_{1_A} (supporting mistreatment of individual women) = 0.39; P_{2_A} (supporting systemic discrimination against women as a group) = 0.45; N_A (neutral) = 0.12;

Hence, the value will be presented as

$$\langle T_{1_A}, T_{2_A}, D_{1_A}, D_{2_A}, D_{3_A}, A_{1_A}, A_{2_A}, A_{3_A}, A_{4_A}, P_{1_A}, P_{2_A}, N_A \rangle = \langle 0.47, 0.35, 0.77, 0.72, 0.77, 0.68, 0.63, 0.41, 0.49, 0.39, 0.45, 0.12 \rangle$$

From the membership values, it can be clearly seen that two or more membership values can be the highest and the same. This kind of capturing is totally not possible while utilizing representation by types of sets. The significant advantages of these refined neutrosophic sets are that they can be converted to less refined sets, like the IRNSs (with twelve membership functions), which can be converted to TRINSs (with five membership functions), which in turn can be converted to SVNS (with three membership functions). Also, we are able to capture extremely granular data representation because we are using these refined neutrosophic sets; this might not be possible with simple fuzzy sets or SVNS. These subtle forms of online sexism can be captured and detected only while we use neutrosophic sets.

Now, we proceed to define the neuro-neutrosophic networks.

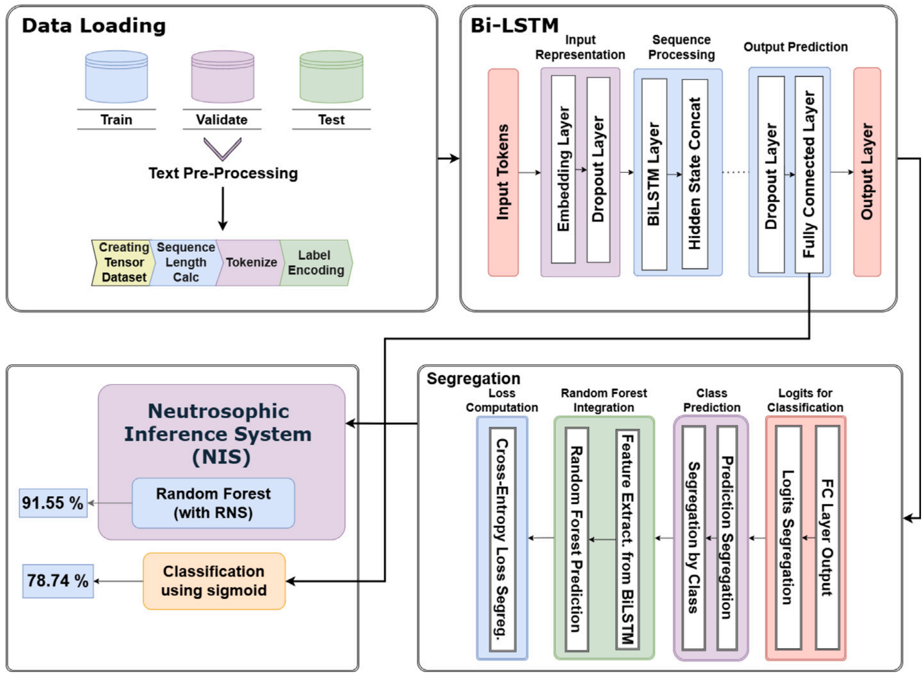

Fig. 5. The system architecture of the deep neuro-neutrosophic system that is based on BiLSTM and then integrated with the NIS-based random forest for the classification task of subtask B.

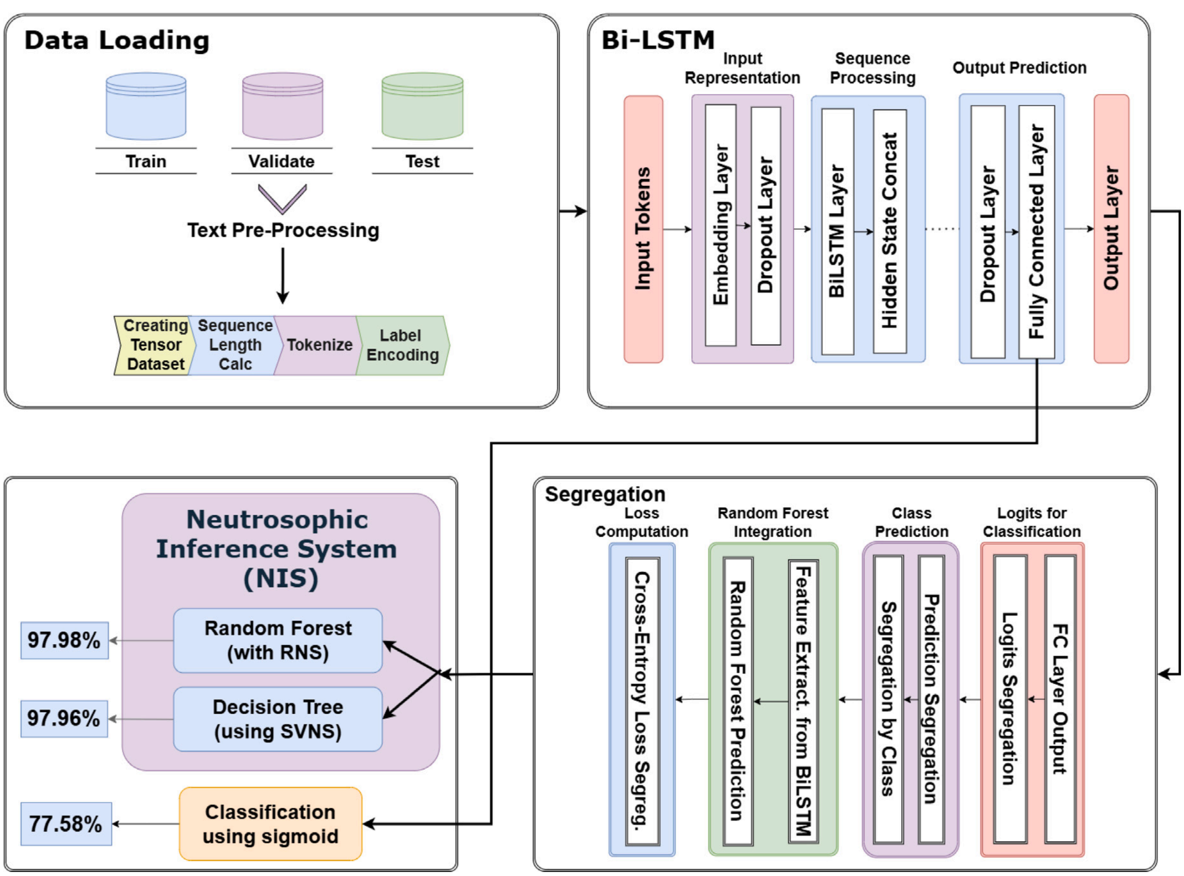

Fig. 6. The system architecture of the deep neuro-neutrosophic system that is based on BiLSTM and then integrated with the NIS-based random forest or NIS-based random forest for the classification task of subtask C.

Table 1
Baselines values for subtasks A, B and C of SemEval Task 10 from Kirk et al. (2023).

Baseline	Model	Continued Pretraining	Macro F1 Subtask A	Subtask B	Subtask C
B0	MostFrequent	No	0.4310	0.1594	0.0317
B1	Uniform	No	0.4509	0.2413	0.0629
B2	XGBoost	No	0.4933	0.2297	0.0881
B3	DistilBERT	No	0.7621	0.5531	0.2935
B4	DistilBERT	Yes	0.7804	0.5367	0.3140
B5	DeBERTa-v3-base	No	0.8235	0.4790	0.1517
B6	DeBERTa-v3-base	Yes	**0.8235**	**0.5926**	**0.3171**

Table 2
Best performing teams on SemEval2023 Task 10: Online sexism detection.

Team name	Model	Subtask A	Subtask B	Subtask C
PingAnLifeInsurance	DeBERTa-v3-large, twHIN-BERT-large	0.8746		
Stce	RoBERTa-large, ELECTRA	**0.8740**	0.7203	0.5487
FiRC-NLP	DeBERTa (ensemble)	0.8740		
JUAGE	PaLM (ensemble)		**0.7326**	
PASSTeam	RoBERTa, HateBERT		0.7212	0.5412
PALI	DeBERTa, RoBERTa			**0.5606**

Table 3
Results for subtask A: The performance of the RoBERTa model and the proposed system (RoBERTa integrated with NIS-based decision tree) in training, validation, and testing dataset of subtask A in terms of accuracy, precision, and recall.

Model	Metric	Train	Validate	Test
	Accuracy	93.9	87.8	91.6
RoBERTA	Precision	89.6	85.4	90.6
	Recall	92.1	86.0	91.2
RoBERTA	Accuracy	98.1	94.7	93.1
+ NIS	Precision	97.9	94.7	93.0
Decision Tree	Recall	98.0	94.7	93.1

3.2. Neuro-neutrosophic networks as a generalization of neuro-fuzzy networks

Neuro-neutrosophy-based classification leverages integrating neutrosophy with neural network models, such as ANNs, CNNs, RNNs, or DNNs, to perform intricate classification tasks. Neuro-neutrosophic systems are a generalization of neuro-fuzzy systems; they are a combination of neutrosophy with neural networks, providing a hybrid technique for modeling and decision-making tasks. Fuzzy systems and DNNs are combined in various ways, as detailed in Talpur et al. (2023). A sequential design is used in two ways: Either by processing the data's features as fuzzy sets that the DNN uses or by the reverse. Similarly, in the reverse process, the DNN's outputs are input for the fuzzy system (Vieira et al., 2004; Mishra et al., 2025). This sequential design of DNN, followed by inference systems, is used in our proposed framework.

Neutrosophic logic functions on the inference of linguistic rules and variables, enabling the articulation of ambiguous, uncertain, or imprecise data in a structured format using neutrosophic membership functions. Conversely, neural networks work on learning relationships and patterns, permitting adaptive and data-driven decision-making. Neuro-neutrosophic systems integrate these two paradigms, utilizing neutrosophic logic's interpretability and the neural networks' learning capabilities to improve system performance. The adaptability of the neuro-neutrosophic systems guarantees that they can be used for diverse real-world problems, proving their versatility.

In the proposed simple neuro-neutrosophic system, two principal components work together: the NIS and a neural network-based model. Powered by neutrosophic logic, the NIS analyzes the input and makes neutrosophic inferences based on domain-specific heuristics or expert knowledge. The neural network then processes the NIS's output, which is represented as neutrosophic or refined neutrosophic sets. The neural network learns from the neutrosophic outputs and refines its inner parameters, enhancing its decision-making capabilities. The combined process, where neutrosophic sets and neural networks converge, is an essential resilience of neuro-neutrosophic systems.

The deep neuro-neutrosophic system extends the basic neuro-neutrosophic framework by incorporating multiple layers of abstraction and complexity. Deep architectures aid in hierarchical feature representation and abstraction, allowing it to learn complicated patterns and relationships.

Several layers of neutrosophic inference and neural networks can be arranged alternatively in a deep neuro-neutrosophic system. Each contributes to the systems' performance and decision-making capacity. The hierarchy allows the proposed system to handle real-world data and tasks requiring nuanced reasoning and decision-making. The deep neuro-neutrosophic systems' architecture has sequential structural designs. Figs. 2 and 3 illustrate both possible cases. In Fig. 2, the NIS is powered by neutrosophy to handle the input, following which the DNN is utilized.

For Fig. 2 the NIS's output can be represented as:

$$y = \sum_{i=1}^{n} w_i f_i(x) \tag{2}$$

where output is y, the weights w_i are assigned by neutrosophic rule and the firing strength $f_i(x)$ of each neutrosophic rule is based on x.

Weight adjustments during learning can be represented as:

$$w_i \leftarrow w_i - \eta \frac{\partial E}{\partial w_i} \tag{3}$$

Consider a simple deep neuro-neutrosophic system that can be used to classify chest X-ray images as COVID-19 or not in disease prediction. Now, the system can be altered to introduce a neutrosophic system to the input and then apply DNN for classification. Each input or pixel can be represented as an SVNS or some other neutrosophic set and then it can be processed by the DNN.

Similarly, as shown in Fig. 3, the DNN's output is used by the NIS for classification. While considering Fig. 3, the output of the DNN integrates with neutrosophic logic as follows:

$$z = f(W_d x + b_d) \tag{4}$$

where z is the output and DNN's weights is given by W_d and biases by b_d. The combined output of the system is represented as:

$$y = g(z, f(x)) \tag{5}$$

where $g(z, f(x))$ is a function that combines the output of DNN with neutrosophic outputs. In this proposed architecture $g(z, f(x))$, which will map the output of DNN to a neutrosophic set with various membership functions. The examples of SVNSs, TRINSs, and IRNSs given previously are example membership values that are obtained from the DNN. These neutrosophic sets are used to create the decision tree or random forest, and according to the rules of the algorithm, every input is assigned a particular class.

4. Methodology

4.1. System architecture

4.1.1. Subtask A

The proposed system architecture involves multiple stages, as illustrated in Fig. 4.

Pre-processing module: The system begins with data loading, where the dataset is divided into three sets: a training set for model training, a validation set used during the training process to tune model parameters and prevent overfitting, and a test set employed post-training to evaluate the model's performance on the unseen data.

DNN module: Once pre-processed, the data is fed into a RoBERTa-base model, which serves as the core of the classification system. The pre-processed text is tokenized and converted into embeddings that capture content and positional information within the sequence.

NIS module: The output of these additional layers is then fed into the NIS, where data is represented as SVNS to the NIS-based decision tree, which categorizes the input into one of three classes: “Sexist” indicating the presence of sexist content, “Non-Sexist” indicating the absence of such content, and “Neutral”, indicating the neutral or ambiguity.

The algorithm of subtask A is illustrated for Algorithm 1.

Algorithm 1 Deep Neuro Neutrosophic Algorithm for Subtask A

Input: Subtask A Dataset $\mathcal{D}$ containing online comments.
Output: Class label $y \in \{\text{Sexist}, \text{Non-Sexist}, \text{Neutral}\}$
Split $\mathcal{D}$ into training set $\mathcal{D}_{train}$, validation set $\mathcal{D}_{val}$, and test set $\mathcal{D}_{test}$.
{**Preprocessing Module**}
for all comment $c \in \mathcal{D}$ **do**
  Remove URLs, special characters, and unnecessary punctuation.
  Expand chat abbreviations using a custom dictionary.
  Replace emoticons with corresponding sentiment text.
  Convert emojis to text using the `emoji` library.
  Normalize and tokenize text using Ekphrasis.
end for
{**Deep Neural Network Module**}
for all preprocessed comment c' **do**
  Encode c' t(`input_ids`, `attention_mask`) using RoBERTa tokenizer
  Pass the encoded input into pre-trained RoBERTa-base model to obtain contextual embeddings.
  Feed embeddings through fully connected layers:
    Dense layer (128 → 32) with ELU activation, L2 regularization
    Final output using sigmoid activation for binary offense score
end for
{**NIS module**}
for all offense score s from model output **do**
  Convert score s to Single Valued Neutrosophic Set (SVNS) representation: (T, I, F)
  Input SVNS into NIS-based decision tree classifier
  Assign final label $y \in \{\text{Sexist}, \text{Non-Sexist}, \text{Neutral}\}$ using decision rules
end for
return Class label y

4.1.2. Subtask B

Fig. 5 illustrates the proposed system architecture.

Preprocessing module: The process starts with loading the data set, which comprises training, validation, and testing subsets. Validation will then be applied during the training process in order to monitor the model’s performance. Following loading, the text has several pre-processing steps. One method is label encoding: the categorical classes, for example, ‘sexist’ or ‘non-sexist’, are given numerical values. Tokenization consists of breaking down sentences into single tokens, which can be thought of in essence as words or even more minor meaningful units.

DNN module: The layers of the Bi-LSTM architecture can be divided into three working groups: the input representation group, the sequence processing group, and the output prediction group. The combined hidden states are then passed on to the fully connected layer, mapping these rich hidden representations onto the final output space. A fully connected layer produces logits, which are not normalized probabilities for each class.

NIS Module: Then, these logits are taken as inputs for the NIS-based random forest, in which they are represented as TRINS with five memberships. Each input will be represented as Threats $T_R(t)$, derogation $D_R(t)$, Animosity $A_R(t)$, Prejudice $P_R(t)$, and Neural $N_R(t)$, membership functions to make the TRINSs. They are used to make the final prediction that will classify the input text into respective categories.

This step starts with computing loss by adopting a cross-entropy loss, which measures the difference between the predictions and true labels. Then, an NIS-based random forest classifier is embedded for further processing with the architecture of the features extracted by Bi-LSTM. The random forest model adds more predictions by combining the results from multiple decision trees to the classification process, thereby adding an extra layer of robustness to the classification process.

The final output of the deep neuro-neutrosophic system (BiLSTM + NIS-based random forest) decides what does or does not belong in one particular class with further subcategories (‘sexist’); an architecture is designed so that the sexist content breaks into clear categories such as threats, derogation, animosity, and prejudiced discussions. These subcategories, therefore, render more precision regarding the nature of sexist feelings, thereby enhancing the overall utility of the model in identifying and differentiating negative behavior types through text. Each layer refines the text data and predictions so that a final classification is both accurate and informative.

The algorithm for Subtask B is illustrated in Algorithm 2.

Algorithm 2 Deep Neuro-Neutrosophic Algorithm for Subtask B

Input: Raw dataset $\mathcal{D}$ with labeled sexist categories
Output: Class label $y \in \{\text{Threats}, \text{Derogation}, \text{Animosity}, \text{Prejudice}, \text{Neutral}\}$
Split $\mathcal{D}$ into training set $\mathcal{D}_{train}$, validation set $\mathcal{D}_{val}$, and test set $\mathcal{D}_{test}$
Encode class labels using a predefined dictionary via `encode_labels()`
for all dataset $\mathcal{D}_i \in \{\mathcal{D}_{train}, \mathcal{D}_{val}, \mathcal{D}_{test}\}$ **do**
  Tokenize text using RoBERTa tokenizer:
    (a) Convert to token IDs
    (b) Generate attention masks
    (c) Pad/truncate sequences to max length 256
  Convert token IDs, masks, and labels to tensors
  Compute sequence lengths from attention masks
  Wrap data into `TensorDataset` and `DataLoader`
end for
Initialize BiLSTM model:
  (a) Embedding layer with output dimension 768
  (b) BiLSTM with 2 layers, hidden size 128, dropout
  (c) Two fully connected layers with ReLU, batch norm
  (d) Final output layer with 5 sigmoid logits
for all batch b in training DataLoader **do**
  Forward pass input through BiLSTM to obtain logits
  Compute cross-entropy loss
  Backpropagate and optimize using Adam optimizer
  Update learning rate using step scheduler
end for
for all logit output z from BiLSTM model **do**
  Convert logits to TRINS representation:
    $(T_R(t), D_R(t), A_R(t), P_R(t), N_R(t))$
  Input TRINS into NIS-based random forest classifier
  Predict class label y using NIS-based decision rules
end for
return Predicted label y for each input

4.1.3. Subtask C

The architecture of subtask C is almost identical to the one applied in subtask B, except that in this model, the output layer is where the results are given a more detailed classification of sexist language. Similar to the previous system used for subtask B, here, two systems are proposed. The first is the BiLSTM model, followed by the NIS-based random forest, and the next is the BiLSTM + NIS-based decision tree (see Fig. 6). The three subtasks are so designed that the first subtask A, determines whether the text has sexist content and, after determining whether it is indeed sexist and belongs to the above-mentioned subcategories (subtask B), further divides it into subcategories (subtask C).

The output classification at a detailed level helps understand all the forms of sexism present in the input, thereby enhancing the utility of the model in content moderation or sentiment analysis.

4.2. Task wise description

The tasks have been divided into three subtasks: A, B, and C. Each of these subtasks is for dividing the comment into various levels to understand the comment. Subtask A contains the division of whether the comment is sexist or not. Since neutrosophy is included in the work, another possibility of neutral comment is added. Since sexism is seen differently by different individuals, it adds a level of uncertainty where some might find it sexist, and some individuals might not. This is where the SVNS is utilized. Adding another category that includes neutral comments increases the accuracy of the detection. So, under subtask A, it is categorized into sexist, not sexist, and neutral.

Subtask B takes the sexist comments that have been detected in subtask A and then further divides them into the types of sexist comments. This subtask includes the four divisions as 1. Threats, plans to harm and incitement, 2. Derogation, 3. Animosity, and 4. Prejudiced Discussion and 5. Neutral. Now, once a comment has fallen into one of these five categories, the processing for subtask C is started. Subtask C has another level of categorization, which provides a clear final description of the comment and its understanding. These subclassifications have already been discussed in the previous sections and are also available in Fig. 1.

4.3. Data set description

This dataset is constructed to support research in detecting and analyzing sexist language within textual content. It is divided into three distinct subsets: training, testing, and validation. These subsets are intended to facilitate model training, evaluation, and validation processes, respectively. The dataset is comprehensively labeled to aid in identifying not only the presence of sexism but also the specific category and sub-category of sexist content.

The data has been collected from two prominent online platforms: Reddit and Gab. These platforms were chosen due to their diverse user bases and varying norms of discourse, providing a broad spectrum of examples of sexist language. By including data from these sources, the dataset captures a wide range of sexist expressions and contexts.

The dataset comprises a total of 22,000 records allocated across three subsets: The training set contains 16,000 records. The validation set consists of 4000 records. This subset is used to fine-tune models and prevent overfitting. The testing set includes 2000 records. This subset is used to evaluate the performance of the trained models.

Each record in the dataset includes five fields: “rewire_id”: A unique identifier for each record, ensuring that each piece of data can be individually referenced and tracked.

“text”: The actual text content that is being analyzed for sexist language.

“label_sexist”: A binary label indicating whether the text is sexist (0) or not sexist (1).

“label_category”: This field specifies the broad category of sexism if the text is labeled as sexist.

“label_vector”: A detailed sub-category label that provides more granular information about the type of sexist content, further refining the categorization provided by the “label category”.

4.4. Data pre-processing

In subtask A, an extensive data pre-processing pipeline is used to clean and standardize text data. It starts by importing essential libraries and configuring Ekphrasis for social media text processing. Training, validation, and test datasets are then loaded. The pipeline removes URLs, converts chat abbreviations using a custom dictionary, and replaces emoticons with their textual equivalents. Emojis are transformed into text descriptions via the emoji library. The text is subsequently normalized and tokenized using the Ekphrasis pre-processor, and unnecessary punctuation is removed. This comprehensive pre-processing ensures the text data is clean and standardized for model training.

For subtask B and subtask C, the data pre-processing pipeline prepares text data for model training using the RoBERTa tokenizer. Initially, text labels are encoded into integer values via a predefined dictionary in the encode_labels function. The load_and_prepare_data function loads the training, validation, and test datasets, applying label encoding to the relevant columns — ‘label_category’ for subtask B and ‘label_vector’ for subtask C — while removing rows with missing encoded labels. The RoBERTa tokenizer then tokenizes the text data, converting each text into token IDs and attention masks, ensuring sequences are padded or truncated to a maximum length of 256 tokens. The tokenized components are concatenated into tensors, and labels are converted into integer tensors. Sequence lengths are calculated from the attention masks for dynamic padding. Finally, a TensorDataset is created containing the input IDs, attention masks, labels, and sequence lengths, and is wrapped in a DataLoader for batch processing and shuffling during model training. The pre-trained RoBERTa tokenizer ensures consistent tokenization, and DataLoaders for the training, validation, and test datasets are constructed using the processed data.

4.5. Module wise description

Data loading and preprocessing module: In data loading, various comments (sourced from Twitter) are loaded for training, validation, and test splits. These are used in the various stages of the model training and testing. The loaded data is preprocessed to standardize the text, allowing the model to train itself with clean and consistent data. Steps carried out to do so include:

1. Removing URLs.
2. Converting chat abbreviations using a custom dictionary.
3. Replacing emoticons with their textual meanings and converting emojis into text descriptions.
4. Using Ekphrasis for normalization and tokenization.
5. Removing unnecessary punctuations.

Model Training module: In this experimental system, the pre-trained transformer RoBERTa model is used for text classification and sentiment analysis tasks in subtask A. RoBERTa was selected due to its exceptional ability to handle contextual subtleties in text, making it a robust and reliable choice for supporting the system’s efforts to incorporate bias handling effectively. The RoBERTa model was further integrated with NIS-based decision tree to create the deep neuro-neutrosophic system for handling subtask A.

A BiLSTM model is implemented for multi-class text categorization for subtasks B and C. The choice of architecture was contingent on the bi-directional nature of the BiLSTM, as it allows the model to address sequential input and capture such relationships in the text from past and future contexts. In subtask B, the BiLSTM model was further integrated with NIS-based random forest to create the deep neuro-neutrosophic system.

In subtask B, the BiLSTM model classifies text into numerous sexist categories based on predefined labels; however, subtask C classifies comments into further subcategories within those labels. The model is initialized with particular embedding and hidden dimensions, and the vocabulary size is calculated using the tokenized dataset.

Further, in subtask C, two deep neuro-neutrosophic systems are proposed: first, the BiLSTM is integrated with NIS-based random forest, and second, the BiLSTM is integrated with a NIS-based decision tree.

Model Configuration: In subtask A, the model utilizes pre-trained RoBERTa transformer architecture to perform offense detection tasks. The input includes two tensors: inputs, which denote token indices, and input masks, which denote whether a token exists in a sequence. The model's architecture is derived from the RoBERTa model, which generates embeddings from the input tensors. After that, the sequence of the following layers is comprised of fully connected (dense) layers with ELU activation and L2 weight decay for reducing overfitting, reducing from 128 to 32 dimensionality, and to one final output layer. This final layer applies sigmoid activation in order to provide offense classification categories. The construction balances structural complexity and runtime efficiencies, leveraging the contextual features associated with the RoBERTa model while creating a concise architecture. The BiLSTM model (for subtasks B and C) consists of three primary blocks: an embedding layer, a fully bidirectional LSTM block with two layers, and fully connected layers. The embedding layer converts from token indices to a dense vector with 768-dimension outputs. The BiLSTM component helps identify contextual linkages by processing the input sequences in both directions, has a hidden dimension of 128, and dropout to mitigate overfitting. After this layer, two fully connected layers resolve to a reduced output of 128, integrating ReLU activation and batch normalization before outputting the logits for 12 classes using a sigmoid activation function, and these logits are used to create refined neutrosophic sets. The model also uses cross-entropy loss for improved convergence, Adam optimizer (learning rate of 0.001), and a step learning rate scheduler, given all this architecture is sufficient to classify sexist comments for both subtasks, without the use of pre-trained transformer models such as RoBERTa.

5. Experimental results and discussions

5.1. Existing baseline/benchmarks

Seven baseline models are supplied to benchmark from simple to complex models as tabulated in Table 1. The easiest baselines involve predicting the most frequent class (B0) and predicting each class (B1). One traditional machine learning model (B2) uses TFIDF vectorization followed by an XGBoost model.

The other baselines use DistilBERT and DeBERTa-v3-base, with fine-tuning on the training set (B3, B5) and continued pre-training (B4, B6) (using 2M unlabeled entries from Gab and Reddit). DeBERTa, with continued pre-training, sets the best baseline across all tasks, with the best performance for subtask A (F1 = 0.8235), while subtasks B and C present more challenges (F1 = 0.5926; 0.3171), with varying degrees of difficulty and potential for performance improvements.

The best-performing teams on the SemEval2023 task 10 are tabulated in Table 2. The team PingAnLifeInsurance (Jin et al., 2023) ranked first for their approach to subtask A, achieving an accuracy of 0.8746. Their system focuses on sentiment analysis for low-resource African languages using Twitter data. It employs a multi-model fusion strategy, incorporating monolingual and multilingual training, translation-based techniques, and a weighted voting method to enhance classification accuracy. Their approach achieved top rankings in multiple languages, including Yoruba, Twi, Nigerian Pidgin, and Swahili, securing second place in the multilingual subtask.

Team FiRC-NLP's (Hassan et al., 2023) approach to SemEval-2023 Task 10 focuses on detecting and explaining online sexism using fine-grained classification. The system utilizes an ensemble of fine-tuned DeBERTa models combined with k-fold cross-validation to improve classification accuracy. The FiRC model achieved second place in subtask A, seventh in subtask B, and fourth in subtask C, demonstrating the effectiveness of their transformer-based ensemble approach for detecting sexism.

JUAGE's (Sorensen et al., 2023) strategy for SemEval-2023 Task 10 involves parameter-efficient tuning for sexism detection. Their model uses BERT fine-tuning for binary classification (subtask A) and prompt-tuning of a 62-billion parameter PaLM model for hierarchical classification (subtasks B and C). By exploiting soft prompt learning with pseudo-tokens and ensemble models, JUAGE attained benchmark rankings by placing 1st in Subtask B with an accuracy of 0.7326. Their approach optimized classification with only a few trainable parameters, demonstrating the effectiveness of prompt-tuning for large pre-trained models at fine-grained sexism detection.

5.2. Subtask A

The RoBERTa model was trained over six epochs, and its accuracy consistently improved from 0.7855 in the first to 0.9335 in the last epoch. The model's training loss decreased from 0.3214 in the first epoch to 0.2083 in the final. Validation accuracy was stable across epochs, fluctuating between 0.8357 and 0.878, while validation loss ranged between 0.4744 and 0.5319. Despite these fluctuations, the model demonstrated consistent generalization performance (see Table 3).

The F1-score for both classes (sexist and non-sexist) was observed to improve over epochs. By the last epoch, the F1-score for classifying sexist comments was 0.726, and for non-sexist comments, it was 0.922, leading to an overall accuracy of 87.8% on the validation set. On the test set, the RoBERTa model attained an accuracy of 91.6%. The difference in F1-scores for classifying sexist and non-sexist comments is due to class imbalance. This can be mitigated with the use of resampling techniques and class weights. The proposed system makes use of class weights to minimize the class imbalance. *Hyperparameters Tuning*: The hyperparameters used for the RoBERTa model given in Table 4.

Deep neuro-neutrosophic system for subtask A:

A pre-trained RoBERTa model and NIS-based decision tree classifiers were used to classify comments for subtask A. The NIS was implemented using SVNS values obtained from the RoBERTa model. These SVNS values were used to train the decision tree classifier. The decision tree classifier was trained on SVNS values and showed excellent performance, with a validation accuracy of 0.981 and a test accuracy of 0.947. On evaluating the test set, the overall test accuracy was 93.1%. The overall test results were consistent with the validation performance.

These results demonstrate that integrating NIS with SVNS values with traditional classification techniques, such as decision trees, can significantly enhance the performance of sexist comment detection, making the models more reliable in classifying sexist and non-sexist content.

Hyperparameters Tuning: In this subtask, the decision tree classifier uses the following hyperparameters: a maximum depth of 10, a minimum of 2 samples required to split a node, and a minimum of 1 sample per leaf node. These parameters ensured a balance between model complexity and overfitting prevention in the binary classification task.

5.3. Subtask B

The BiLSTM model showed a consistent improvement in accuracy throughout the training process, starting with an initial accuracy of 0.8150 in the first epoch and gradually increasing to 0.8359 by the final epoch. This steady increase suggests that the model effectively learned from the training data. The corresponding training loss exhibited a downward trend. However, the validation loss did not decrease as consistently, showing some fluctuations over the epochs. Despite this, the validation accuracy remained relatively stable, hovering around 0.7912. This stability indicates that the model can generalize reasonably well to the validation data, maintaining a consistent level of performance even when encountering new data (see Table 5).

Table 4
Hyperparameter tuning for RoBERTa model used for subtask A.

Parameter	Range	Optimal value
Transformer model	['bert-base', 'roberta-base']	Roberta-base
Input sequence length	[64, 100, 128]	100
Dropout rate	[0.1, 0.3, 0.5]	0.1
Dense layer units	[64, 128, 256]	128, 32, 3, 32, 128
Activation functions (Dense layers)	['elu', 'relu', 'gelu']	Elu
Batch normalization	[Yes, No]	Yes
Optimizer	['Adam', 'SGD', 'RMSProp']	Adam
Learning rate	[1e−5, 3e−5, 1e−4]	3e−5
Loss function	['BinaryCrossentropy', 'MSE']	BinaryCrossentropy
Metrics	['accuracy', 'precision', 'recall']	Accuracy
Output activation	['sigmoid', 'softmax']	Sigmoid
Output units	[1, 2] (binary vs. multi-class)	1
Pretrained layer trainable	[True, False]	True (default)

Table 5
Results for Subtask B: The performance of the BiLSTM model and the proposed deep neuro-neutrosophic model (BiLSTM integrated with NIS-based random forest) in training, validation, and testing dataset of subtask B in terms of accuracy, precision, and recall.

Model	Metrics	Train	Validate	Test
BiLSTM	Accuracy	83.59	79.12	78.74
	Precision	83.26	76.45	76.53
	Recall	83.02	75.95	77.25
BiLSTM + NIS random forest	Accuracy	91.81	91.37	91.55
	Precision	94.23	86.16	87.16
	Recall	87.75	86.48	85.45

Hyperparameters Tuning: The hyperparameters used for the RoBERTa model given in Table 6.

Deep neuro-neutrosophic system for subtask B:
An NIS-based random forest classifier was trained on the features generated by the BiLSTM model to improve performance. The proposed system achieved a validation accuracy of 0.9137, which is improved compared to the accuracy of the BiLSTM model of 0.8359.

The proposed model achieved an accuracy of 0.9155 on the test set. This performance is closely aligned with the validation accuracy, which hovered around 0.9137. The consistency between validation and test accuracies further shows the model's ability to generalize well to unseen data.

The result of 91.55% accuracy on the test set suggests that the model maintained its robustness across different datasets, including new, unseen data. This indicates that the model is not overfitting and can be considered reliable for practical classification tasks in real-world scenarios. The relatively high accuracy also implies that the chosen architecture and training process were effective for the given task.

Hyperparameters Tuning: For subtask B, a random forest classifier is employed with 200 estimators (trees) to improve classification performance, a maximum tree depth of 15 to capture complex patterns, a minimum of 3 samples required to split a node, and a minimum of 2 samples per leaf node to enhance generalization. This setup helps effectively classify comments based on their severity levels.

5.4. Subtask C

The experimental results show the effectiveness and robustness of the BiLSTM classification model for subtask C. It is further improved while using an NIS along with the BiLSTM classification. The results from the training, validation, and test phases are tabulated in Table 7.

The BiLSTM model showed a consistent improvement in training accuracy across the five epochs, starting from an accuracy of 0.8059 and increasing to 0.8157 by epoch 5. This steady increase suggests the model learned the patterns and features necessary for classification from the training data. The training loss decreased over the epochs, indicating that the model successfully minimized the error during the training process. However, the validation loss did not exhibit the same consistent downward trend, fluctuating slightly over the epochs. This fluctuation could be due to the complexity of the validation data or potential overfitting. Despite these fluctuations, the validation accuracy remained relatively stable, with values around 0.7961.

Deep neuro-neutrosophic systems for subtask C:
NIS-based decision tree classifier: The validation accuracy achieved using the decision tree classifier was 97.97%, which is improved compared to the validation accuracy of the BiLSTM model alone. The increased performance indicates that the NIS-based decision tree can capture additional patterns in the data and significantly improve the performance of the BiLSTM model.

*NIS-based random forest classifier:*The neutrosophic set-based random forest classifier also showed a validation accuracy of 97.97%. This consistency with the decision tree suggests that ensemble methods provided a significant performance boost.

The final evaluation of the test dataset yielded an accuracy of 0.7877 without using decision trees or a random forest classifier. A higher accuracy of 97.96 and 97.98 is observed using decision trees and random forest classifiers, respectively. This test accuracy, close to the validation accuracy, indicates that the model generalizes well to unseen data. The slight drop in performance from the validation to the test set is expected and typical, reflecting the model's robust learning and capability to handle real-world data.

Hyperparameters Tuning: Both random forest and decision tree classifiers are used in subtask C. The random forest classifier consists of 300 estimators, with a maximum depth of 20, a minimum of 4 samples needed to split a node, and a minimum of 2 samples per leaf node, allowing the model to handle the complexity of multi-label classification across 12 categories. The decision tree classifier follows a similar configuration with a maximum depth of 20, a minimum of 4 samples required for splitting, and two samples per leaf node, ensuring effective decision boundaries in multi-label classification.

Hyperparameter tuning: Hyperparameter tuning process was carried out to refine the model's architecture. The hyperparameters adjusted were the learning rate, dropout rate, and number of dense units. Various learning rates were evaluated, and 3e−5 emerged as the most stable and effective. This value ensured gradual yet consistent convergence, allowing the model to better learn subtle patterns in imbalanced datasets without overshooting the minima. After testing various levels of regularization, a dropout rate of 0.3 was chosen; it successfully reduced overfitting by randomly deactivating nodes during training, which is especially useful when learning from a skewed class distribution because it forces the model to generalize better rather than memorize dominant class patterns. Furthermore, using 256 dense units in the first fully linked layer enabled the model to preserve rich feature representations while preserving training efficiency. These tuning efforts resulted in significant gains in validation accuracy, as the model was able to better identify between 'sexist', 'non-sexist', and 'neutral' classes. The hyperparameters of subtasks B and C are given in Table 6 and of subtask A in Table 4. The improved configuration also helped the model respond more evenly to all classes, making it better

Table 6
Hyperparameter tuning for BiLSTM used in Subtask B and Subtask C.

Parameter	Range	Optimal value
Embedding dimension	[64, 128, 256]	128
Hidden dimension	[128, 256, 512]	256
Number of output labels	[2, 3, 5, 6]	5
Bidirectional LSTM	[True, False]	True
Number of LSTM layers	[1, 2, 3]	2
LSTM dropout	[0.3, 0.5, 0.7]	0.5
Batch size	[16, 32, 64]	32
Activation functions	['ReLU', 'GeLU', 'ELU']	ReLU
Output activation function	['Sigmoid', 'Softmax']	Sigmoid (for multi-label)
First dense layer units	[64, 128, 256]	128
Second dense layer units	Equal to number of labels	5
Batch normalization	[Yes, No]	Yes
Final output shape	[num_labels]	[5]
Packed sequence usage	[Yes, No]	Yes
Optimizer (External to class)	['Adam', 'SGD', 'RMSProp']	Adam (commonly used)
Loss function (External to class)	['BCELoss', 'BCEWithLogitsLoss']	BCELoss

Table 7
Results for Subtask C: The performance of the BiLSTM model and the proposed deep neuro-neutrosophic model (BiLSTM integrated with NIS-based random forest and BiLSTM integrated with NIS-based decision tree) in training, validation, and testing dataset of subtask C in terms of accuracy, precision, and recall.

Model	Metrics	Train	Validate	Test
BiLSTM	Accuracy	81.57	79.61	78.77
	Precision	92.81	86.72	80.42
	Recall	69.80	71.80	69.70
BiLSTM + NIS decision tree	Accuracy	97.96	97.97	97.96
	Precision	92.70	77.00	65.80
	Recall	35.60	41.00	41.00
BiLSTM + NIS random forest	Accuracy	97.98	97.97	97.98
	Precision	79.19	96.20	58.90
	Recall	48.50	49.70	46.50

at detecting less common or unclear cases—an important feature for handling subjective and context-based content like online sexism.

Representational learning: Throughout all three subtasks, representational learning was mostly powered by pre-trained transformer models, namely RoBERTa, which best captured contextual and semantic nuances of the comments. Rather than being dependent on manual feature engineering, the model learned deep contextual embeddings from raw text inputs. These embeddings were then fed into further dense layers to learn higher-level features that are more suited to classification. Utilization of Ekphrasis for social media task-specific preprocessing, including slang expansion, emoji normalization, and informal text handling, further improved the input representation quality. This enabled learning subtle patterns in the data, particularly beneficial for detecting subtle or implicit sexism. By bridging transformer-based embeddings with properly optimized dense layers and neutrosophic decision logic, the system provided resistant and expressive feature representations to each task.

5.5. Deep neuro-neutrosophic systems on HateXplain dataset

The proposed system was applied to the HateXplain dataset (Mathew et al., 2021). The obtained accuracy while using BERT alone was 70.9%. When the deep neuro-neutrosophic framework combined the BERT model with NIS-based random forest, the obtained accuracy was 77.9%. This improvement highlights the improved performance when integrating NIS-based classifiers with deep learning models in abusive online language detection tasks.

5.6. Implications of the study

The proposed deep neuro-neutrosophic system integrates deep learning methods with NIS; it uses neutrosophic sets' capability to handle bias and uncertainty with the pattern recognition skills of deep learning models. The results tabulated in Table 8 demonstrate the effectiveness of the proposed approach; it can more accurately and reliably detect sexist language than existing deep learning models with neutrosophy.

Table 8
Table of comparison of accuracy of deep learning model and deep neuro-neutrosophic system for all three subtasks.

Model/Subtask	Deep learning model	Proposed deep neuro-neutrosophic model
Subtask A	91.6	**93.1**
Subtask B	78.74	**91.55**
Subtask C	78.77	**97.98**

Theoretical and practical implications: The framework provides an overall framework for combining neutrosophy with DNNs. Our findings align with prior research indicating the effectiveness of neutrosophy in deep learning models in NLP tasks since integrating NIS provides a significant advancement. Previous studies have mainly focused on traditional machine learning models or standard deep learning approaches, which often struggle with ambiguities and uncertainties in human language; they have not studied the combination of neutrosophic models and deep learning. The incorporation of neutrosophy allows the proposed model to account for various degrees of sexism, which enhances its ability to detect subtle sexist content. It is also an improvement from previous works that relied on binary or categorical classifications without accommodating the complexity of neutral or indeterminate states.

The proposed models open the field to several new ways to combine neutrosophic logic and DNNs; these models can also be used to analyze real-world problems involving human factors.

Application in Content Moderation and Social Media Platforms: The proposed system presents a significant potential for integration into content moderation and social media platforms. It can be used to automatically identify a broad range of sexist language, from explicit to ambiguous forms, which makes it highly applicable in environments where ambiguity and bias are common. Automatic detection enables real-time flagging of offensive content, reducing the need for manual review and allowing platforms to respond quicker. Additionally, social media platforms can aid in user behavior monitoring, identifying individuals who repeatedly engage in sexist conduct, and taking appropriate actions such as issuing warnings, temporary bans, or account suspensions.

The model uses neutrosophic logic to identify this broad spectrum of sexist content and aid in context-sensitive moderation. This capability helps strike a balance between mitigating harmful content and avoiding excessive censorship. Adopting intelligent moderation tools can improve platform integrity, safeguard users from dangerous interactions, and help create a more inclusive digital space.

Limitations of the study: While the proposed system demonstrates improved accuracy and nuanced detection of sexist content, there are limitations to consider. The model's performance is still partially constrained by the availability and quality of annotated datasets, especially for subtle and underrepresented forms of sexism. The inherent subjectivity in categorizing sexist language creates possible annotation bias, which may limit the model's generalizability. Furthermore, while neutrosophic logic improves the system's interpretability and sensitivity to uncertainty, it also adds complexity that could constrain scalability and interpretability for non-technical stakeholders. While neutrosophic logic introduces interpretability and resilience to uncertain or ambiguous situations, it also makes the model pipeline more complex, requiring more advanced tuning and deployment infrastructure.

Computational Overload: While the suggested deep neuro neutrosophic system is more accurate and reliable in detecting online sexism, it introduces some computational overhead. The model is more complex regarding memory utilization and processing time as it integrates NIS with deep learning architectures. Training and inference costs are incurred due to the need to calculate and recompute truth, indeterminacy, and falsity membership values at many levels of the neural network. In addition, processing neutrosophic representations together with standard features requires more comprehensive data preprocessing and parameter optimization. Such a computational burden can be problematic in real-time deployment applications or on limited-resource hardware. Potential research directions in the future may explore optimization methods, model pruning, or hybrid hardware acceleration methods to trade off performance gains for computation efficiency.

6. Conclusions and future work

The proposed deep neuro-neutrosophic system was used for SemEval 2023 Task 10: Explainable Detection of Online Sexism to predict whether a text is considered sexist and which type it is, handling all three subtasks. The framework provides a more nuanced comprehension of online material by accounting for ambiguity and neutral and indeterminate states in language. The addition of the NIS increased the model's detection accuracy of different forms of sexist language.

Improvement in results: In the testing phase, the system obtained 93.1% accuracy in subtask A, 91.55% in subtask B, and 97.98% in subtask C, compared to DNN models that obtained 91.6% accuracy in subtask A, 78.74% in subtask B, and 78.77% in subtask C.

These results clearly highlight the improvement in accuracy by using NIS in combination with DNN.

Future work can include a more diverse range of data sources and languages to improve the model's generalizability and robustness across different cultural and contextual areas. The model can also be run on real-time datasets, which will be dynamic and helpful in real-life usage. Also, the proposed system can be used for various real-world problems that can be solved using deep learning. Future research can focus on improving the integration of other refined neutrosophic sets with deep-learning models. Exploring different ways to represent and quantify indeterminacy and neutrality in the text could further enhance the deep neuro-neutrosophic systems' performance. Investigating other forms of neutrosophic sets and their applications in different NLP tasks could provide valuable insights.

CRediT authorship contribution statement

Ekta Biju: Methodology, Formal analysis, Data curation. **Ilanthenral Kandasamy:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Samiksha Sai Paladugu:** Software, Methodology, Formal analysis, Data curation. **Ashwati Naduparambil Alakkal Sajit:** Validation, Methodology, Formal analysis, Data curation. **Ratnala Baby Sowmya:** Writing – original draft, Software, Methodology, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data is available online.

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